Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer specializing in LLM orchestration, RAG system design, and agentic workflow deployment using LlamaIndex and LangChain - with production healthcare deployments achieving measurable retrieval and reliability improvements. 2.5+ years building and shipping LLM/RAG/agent applications in regulated environments: ~35% retrieval precision improvement, >30% hallucination reduction, and ~25% agent stability gains. Experienced in identifying production LLM pipeline failure modes and designing structural solutions (schema-validated agent contracts, intermediate transformation layers, grounding-rule frameworks). Master's degree, Kent State University. Passionate about the LlamaIndex ecosystem and committed to translating real-world deployment insights back into framework improvements.

CORE COMPETENCIES

LlamaIndex • LangChain • RAG System Design • Agentic Workflows • LLM Orchestration • Multi-Agent Pipelines • Prompt Engineering • Evaluation Pipelines • Hallucination Mitigation • Vector Search • Semantic Retrieval • Python • FastAPI • Docker • PyTorch • scikit-learn • Healthcare AI (HIPAA) • Production LLM Deployment

PROFESSIONAL EXPERIENCE

Progress Solutions Inc. *Jul 2025 - Present*

Jr. AI/ML Engineer Trainee - Healthcare Technology & AI Consulting

- Designed and deployed **RAG systems for clinical knowledge retrieval** using LlamaIndex and LangChain - implemented recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by **~35%** and reduced irrelevant retrieval by **>30%**. Grounded response alignment exceeded **90%** in evaluation. Identified chunking strategy trade-offs specific to clinical long-form text - insights with direct framework-improvement value.
- Built **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by **~25%**. Applied grounding-rule constraints that outperformed model-specific prompt engineering - a generalizable production LLM pattern.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; improved recall by **15-20%** on high-risk categories while holding precision **>90%**.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by **~30%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage documentation, evaluation audit trails, and system-limitation documentation for stakeholders.

Energy Solutions International (Emerson) *Jun 2022 - Apr 2023*

Database & DevOps Performance Engineer (Intern) - Oil & Gas, Enterprise ERP

- Optimized SQL and **T-SQL stored procedures** for the Synthesis Order-to-Cash platform; reduced query execution time **~20%** and data retrieval latency **~25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors **>30%** and improved release cycle time **~35-40%**.
- Designed performance dashboards using **Grafana** to track long-running queries, deadlocks, and CPU/I/O contention; cut incident recurrence **~25%**.

Suvidha Foundation *Jan 2022 - Mar 2022*

Machine Learning Engineer - Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization and highlight-generation system** across 5,000+ sessions; reduced manual review time by **60-70%**.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs **~30%** and raised contextual answer accuracy **>85%**.

KEY PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents using LlamaIndex; ANN similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces. Production-grade agentic orchestration pattern applicable across regulated industries.

LlamaIndex, LangChain, Python, RAG, multi-agent orchestration, vector search

Manga Lens - Chrome Extension for Real-Time AI Translation shipped

Full-stack browser extension integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local) with provider abstraction layer, per-provider payload optimization, 7-day translation cache, and 29-site selector configurations. Multi-provider architecture demonstrates production LLM integration patterns - provider-agnostic interface design with graceful degradation.

JavaScript (Manifest V3), Claude API, GPT-4o, Ollama, multi-provider LLM integration

Dream Decoder - Multimodal AI Creative Intelligence Platform

Full-stack FastAPI + React app with **intermediate structured prompt transformation layers** between interpretation and generation stages - improved contextual alignment **~30%** and first-pass image success rate **~25-30%** over naive direct-prompt orchestration. Demonstrates LLM pipeline seam engineering - a generalizable production pattern for multi-step LLM applications.

FastAPI, React/TypeScript, Perplexity Sonar, Replicate image models, structured prompt transformation

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN with unified **YOLOv8** model; reduced inference latency **~30%**. Sliding-window confidence aggregation suppressed false positives **~25%**.

YOLOv8, PyTorch, OpenCV, real-time inference optimization

EDUCATION

Master's Degree - *Kent State University, Ohio, USA* *Aug 2023 - May 2025*

TECHNICAL SKILLS

LLM Frameworks: LlamaIndex, LangChain, Hugging Face Transformers, structured outputs, evaluation pipelines, guardrails

AI/ML: RAG system design, agentic workflows, multi-agent orchestration, prompt engineering, vector search, embeddings, semantic retrieval

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL (T-SQL, PostgreSQL)

ML Libraries: scikit-learn, XGBoost, PyTorch, YOLOv8, OpenCV, ControlNet

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, cloud-ready deployment

Domains: Healthcare AI (HIPAA-conscious), Production LLM Deployment, Regulated Industry AI, Applied AI Engineering

Work Authorization: F-1 OPT - authorized to work in the United States now; future sponsorship required